

KEERTHAN NEKKANTI

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EDUCATION

Carnegie Mellon University, Tepper School of Business – Pittsburgh, PA 05/26

Master of Science in Business Analytics - MSBA

- Teaching Assistant for Introduction to R and Python.

Manipal Academy of Higher Education, Manipal Institute of Technology – Manipal, India 07/22

Bachelor of Technology, Computer Science Engineering - Big Data Specialization

- Memberships: Database Head of Sports Club, Information Security Club, Cultural Club

EXPERIENCE

Bank of America – Hyderabad, India 07/22 – 07/25

Officer, Software Engineer 1A (07/24 – 07/25)

- Process Improvement:** Automated data collection from seven markets using Java Spring Boot and React.js, ensuring compliance with international economic sanctions.
- Data Management:** Onboarded and managed 100M+ records of sanctions data, optimizing document framework application for real-time data processing and accuracy.
- Data Processing:** Implemented advanced algorithms and automation framework, reducing processing time from three hours to 10 minutes and decreasing server processing hours by ~24%.
- Initiative:** Established data controls to assess quality, enhancing system scalability by absorbing a 60% increase in volume with zero downtime; Recipient of the BofA Gold Award for outstanding performance.

Officer, Software Engineer 1B (07/22 – 06/24)

- Problem Solving:** Designed a neural network algorithm to process 10M transactions weekly, streamlining detection of money laundering patterns, improving processing speed by 50%, and establishing rapid response capabilities.
- Data Analysis:** Created an NLP model to analyze money laundering patterns, reducing false positives by 20%.
- Innovation:** Created a model to generate synthetic data, facilitating application creation across the globe while complying with regulations; Recipient of the BofA Silver Award for outstanding performance.

PROJECT EXPERIENCE

MapShock – Enterprise-Grade Agentic AI Platform | *Personal Project*

- Agentic AI Orchestration:** Developed an intelligent orchestrator coordinating 46 specialized AI agents to automate complex strategic intelligence gathering across finance, geopolitics, cybersecurity, and supply chain domains.
- Automated Reporting Pipeline:** Engineered a protocol-driven system to synthesize massive datasets into analyst-quality intelligence reports in minutes, dramatically reducing analysis time and operational costs compared to traditional consulting workflows.
- Quality Assurance & Governance:** Implemented a rigorous 6-gate quality pipeline to systematically verify model outputs, ensuring every analytical claim is backed by fully cited, timestamped evidence for enterprise-level reliability.

Deloitte (MSBA Capstone Course Project) – Pittsburgh, PA 01/26 – 04/26

Data Sourcing Lead

- Data Sourcing:** Designed and implemented SQL-based data acquisition and ETL pipelines for efficient data integration and management.
- Data Sanitization:** Performed data cleaning, preprocessing, and quality assurance to ensure accuracy and consistency across datasets.

Performance Analysis of Lightweight Ciphers for Securing Images – Manipal, India 01/22 – 06/22

IEEE Conference Paper

- Analysis:** Conducted a comprehensive analysis of lightweight encryption algorithms, including SIT, LEA, Twine, and Present, evaluating their effectiveness in securing digital images.
- Performance Measurement:** Identified key performance metrics such as encryption speed, security robustness, and computational efficiency, providing insights into optimal ciphers for resource-constrained environments.

Social Interaction Data Capture App for Opportunistic Networking Analysis – Manipal, India

Academic Project

05/21 – 07/21

- Product Development:** Developed an innovative mobile application to capture social interactions, creating a comprehensive dataset to support analysis of opportunistic networking scenarios.

SKILLS

- Software Development:** Java, Python, Spring Boot, React.js, Flutter, Django, RESTful APIs
- Frameworks:** Teradata, TensorFlow, Scikit-learn, Apache Spark, Docker

ADDITIONAL INFORMATION

- Achievements:** Runner-Up in Bank of America Innovation Challenge, 2nd Place in Andhra Pradesh State Abacus Competition
- Certifications:** Neural Networks and Deep Learning, Structuring Machine Learning Projects, Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Portfolio:** github.com/Keerthan28